

Cryptography

CSIT — 5th Semester — 2080 — Answer Sheet
Subject Code: CSC327

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Section A

1. Define discrete logarithms. How key generation, encryption and decryption is done in RSA. In a RSA cryptosystem, given $p=13$ and $q=7$, determine private key, public key, and perform encryption and decryption for the text $M= "hi"$ using 0 to 25 for letters from a to z.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

2. Write down the encryption and decryption process at 2-DES and 3-DES. Explain the Fiestel cipher structure. Divide $5x^2 + 4x + 6$ by $2x + 1$ over GF (7).

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. What are the applications of hash functions? Discuss how SHA-1 algorithm generates hash value from a given message.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

Section B

Attempt any Eight questions[8×5=40]

4. Show encryption and decryption of "csit" using hill cipher having key $K = 3257$

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. Give the formal definition of Authentication system. Describe about one way and mutual Authentication system.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. List the stage of certificate life cycle. What are the types of firewalls?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. What is malicious logic? How Zombies are different from Trojan horses?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. How Miller Rabin test is used for primality testing? Show whether the number 561 passes the test.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. Show that the set of integers is Ring under addition and multiplication.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. How substitution ciphers are different from transposition ciphers? Given a message $M = \text{"CSIT PROGRAM IS A HOT CAKE"}$, encrypt M using Rail Fence cipher with rail size=3.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. Describe about IPsec. List the five services of PGP.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

12. How does meet in middle attack work in Diffie Hellman key Exchange protocol? Explain.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.